FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE (Advanced), 2015

PART TEST – III

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 180**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 10)** contains 10 multiple choice questions which have only one correct answer. Each question carries **+2 marks** for correct answer. There is no negative marking.

Section-A (11 to 15) contains 5 multiple choice questions which have more than one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

(ii) **Section-C (01 to 05)** contains 5 Numerical based questions with answers as numerical value and each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

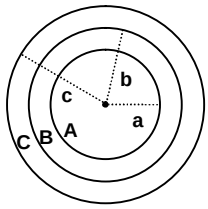
PART – I

SECTION – A

Single Correct Choice Type

*This section contains **10 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.*

1. Three concentric charged metallic spherical shells A, B and C have radii a, b and c; charge densities σ, $-\sigma$ and σ; and potentials V_A, V_B and V_C respectively. Then which of the following relations is correct?



(A) $V_A = \dfrac{(a+b+c)\sigma}{\varepsilon_0}$

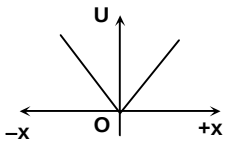
(B) $V_B = \dfrac{[(a^2/b)-b+c]\sigma}{\varepsilon_0}$

(C) $V_C = \dfrac{[(a^2+b^2)/c+c]\sigma}{\varepsilon_0}$

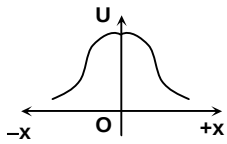
(D) $V_A = V_B = V_C = \dfrac{(a-b+c)\sigma}{\varepsilon_0}$

2. Four identical charges are placed at the four vertices of a square lying in YZ plane. A fifth charge is moved along X axis. The variation of potential energy (U) along X axis is correctly represented by

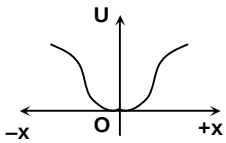
(A)



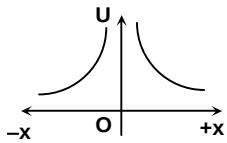
(B)



(C)



(D)

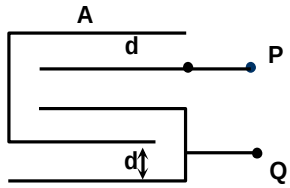


3. Two resistances 300 ohm and 400 ohm are connected in series with a 60 volt supply. A voltmeter connected across 400 ohm reads 30V. The same voltmeter when connected across the other resistance will read.

(A) 22.5 V
(B) 50 V
(C) 30 V
(D) 10 V

Space for rough work

4. The equivalent capacitance between P and Q in the figure is (area of each plate is A and separation between two consecutive plates is d)



(A) $\frac{A\varepsilon_0}{2d}$ (B) $\frac{3A\varepsilon_0}{2d}$

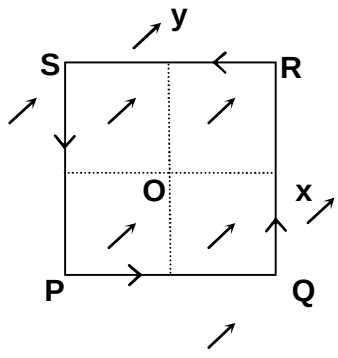
(C) $\frac{2A\varepsilon_0}{d}$ (D) $\frac{5}{3}\frac{A\varepsilon_0}{d}$

5. Two long parallel straight conductors carry currents i_1 and i_2 ($i_1 > i_2$). When the current are in the same direction, the magnetic field at a point midway between the wires is 20 μT. If the direction of i_2 is reversed, the field becomes 50 μT. the ratio of the currents i_1/i_2 is

(A) 5/2 (B) 7/3

(C) 4/3 (D) 5/3

6. A uniform constant magnetic field $\vec{B}$ is directed at an angle of 45° to the axis in the xy –plane. PQRS is a rigid, square wire frame carrying a steady current I_0, with its centre at the origin O. At time t = 0, the frame is at rest in the position as shown in figure, with its sides parallel to the x and y axes. Each side of the frame is of mass M and length L. The torque τ about O acting on the frame due to the magnetic field will be?



(A) $\frac{BI_0L^2}{\sqrt{2}}\left[+\hat{j}-\hat{i}\right]$ (B) $\frac{BI_0L^2}{\sqrt{2}}\left[-\hat{j}+\hat{i}\right]$

(C) $BI_0L^2\left[+\hat{j}-\hat{i}\right]$ (D) $BI_0L^2\left[-\hat{j}+\hat{i}\right]$

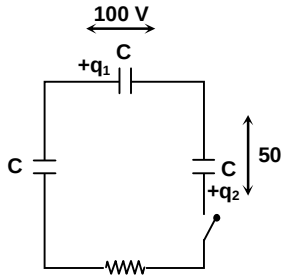
7. A coil is in the form of equilateral triangle of side 'a'. The current in the coil is I. The magnetic field at a distance 'a' equidistant' from all the vertices will be

(A) $\frac{\mu_0 I}{2\sqrt{3}\pi a}$ (B) $\frac{\sqrt{3}\mu_0 I}{2\pi a}$

(C) $\frac{\mu_0 I}{\sqrt{3}\pi a}$ (D) $\frac{\mu_0 I}{4\sqrt{3}\pi a}$

Space for rough work

8. Two capacitor each of capacitance C = 1 μF each charged to potential 100 V and 50 V respectively and connected across a uncharged capacitor 'C' = 1μF as shown in the figure if switch 'S' is closed then charge in the uncharged capacitor at equilibrium will be



(A) $\frac{50}{3}\mu C$ (B) 50 μC

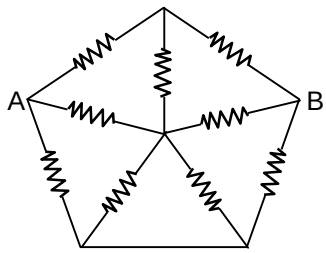
(C) 25 μC (D) $\frac{100}{3}$ μC

9. Three point charges +q, −2q and +q are placed at points $\left(0, \frac{a}{\sqrt{2}}, \frac{a}{\sqrt{2}}\right)$; (0,0,0) and $\left(0, \frac{a}{\sqrt{2}}, -\frac{a}{\sqrt{2}}\right)$ respectively. Find the net electric dipole moment vector of this charge assembly.

(A) $\sqrt{2}qa(-\hat{j})$ (B) $\sqrt{2}qa(-\hat{k})$

(C) $\sqrt{2}qa(+\hat{j})$ (D) $2qa(+\hat{j})$

10. if resistance of each resistor is R, then effective resistance between points A and B of the shown network is



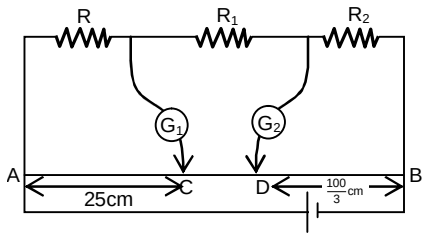
(A) $\frac{8R}{11}$ (B) $\frac{6R}{11}$

(C) $\frac{6R}{5}$ (D) $\frac{2R}{3}$

Multiple Correct Answer(s) Type

*This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.*

11 The diagram show is a modified meter bridge, which is used for measuring two unknown resistances R_1 and R_2 at the same time. When only the first galvanometer is used, balance point is found at point C. Now the first galvanometer is removed and the second galvanometer is used, which gives balance point D. Using the details given in the diagram, find out the value of R_1 and R_2



(A) $R_1 = 5\ R / 3$ (B) $R_2 = 4\ R / 3$

(C) $R_1 = 4\ R / 3$ (D) $R_2 = 5\ R / 3$

Space for rough work

12. 10C of charge is given to a conducting spherical shell and another– 3C point charge is placed inside the shell. For this arrangement mark out the correct statement(s).
(A) The charge on the inner surface of the shell will be +3C and it can be distributed uniformly or non uniformly
(B) The charge on the inner surface of the shell will be +3C and its distribution would be uniform
(C) The net charge on outer surface of the shell will be +7 C and its distribution can be uniform or non uniform
(D) The net charge on outer surface of the shell will be +7C and its distribution would be uniform

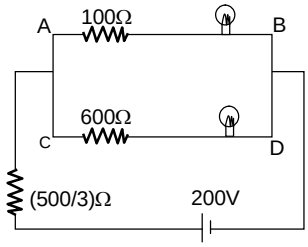
13. A parallel plate capacitor of plate area A and plate separation d is charged to a potential difference V and then the battery is disconnected. A slab of dielectric constant K is then inserted between the plates of the capacitor so as to fill the space between the plates. If Q, E and W denote the magnitude of charge, the electric field between the plates (after the slab is inserted) and magnitude of the work done on the system in question during the process of inserting the slab, then :

(A) $Q = \frac{\varepsilon_0 AV}{d}$ (B) $Q = \frac{\varepsilon_0 KAV}{d}$

(C) $E = \frac{V}{Kd}$ (D) $W = \frac{\varepsilon_0 AV^2}{2d}\left(1-\frac{1}{K}\right)$

14. A proton is fired form origin with velocity $\vec{v} = v_0\hat{j} + v_0\hat{k}$ in a uniform magnetic field $\vec{B} = B_0\hat{j}$. In the subsequent motion of the proton
(A) its z coordinate can never be negative
(B) its x coordinate can never be positive
(C) its x and z coordinates cannot be zero at the same time
(D) its y coordinate will be proportional to its time of flight.

15. Two bulbs 25W, 100V (upper bulb in figure) and 100W, 200V (lower bulb in figure) are connected in the circuit as shown in figure. Choose the correct answer (s)
(A) Heat lost per second in the circuit will be 80J
(B) Ratio of heat produced per second in bulbs will be 1: 1
(C) Ratio of heat produced in branch AB to branch CD will be 1:2
(D) Current drawn from the cell is 0.4 A



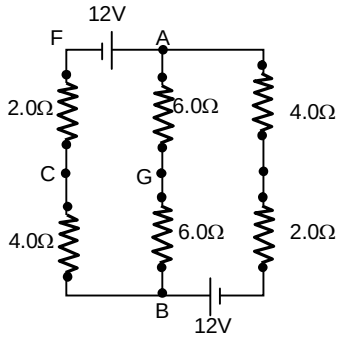
Space for rough work

SECTION – C

(Integer Answer Type)

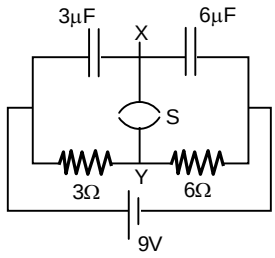
*This section contains **5** questions. The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The bubble corresponding to the correct is to be darkened in the ORS.*

1. In the network shown in the figure, the potential difference (in Volt) across points A and B is

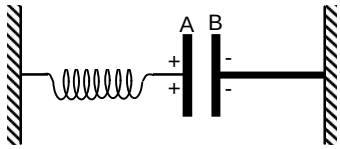


2. Two circular coils of radii 5cm and 10 cm carry equal current of 2A in opposite sense . The coils have 50 and 100 turns respectively and are placed in such a way that they lie in same plane and their centres coincide. Magnitude of magnetic field(in Tesla) at the common centre of coils is

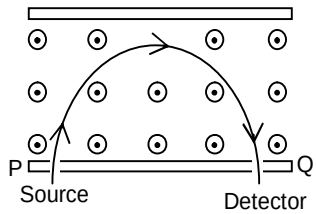
3. A circuit is connected as shown in the figure with the switch S open. When the switch is closed, if total amount of charge that flows from Y to X is Q(in μC), then $\frac{Q}{3} =$



4. Plate A of a parallel plate air filled capacitor is connected to a spring having force constant k and plate B is fixed. They rest on a frictionless table top as shown in the figure. if a charge +q is placed on plate A and a charge –q on plate B, by how much does the spring expand in equilibrium ($q^2 = 6kA\varepsilon_0$)



5. A uniform magnetic field with a slit system as shown in the figure is to be used as a momentum filter for high energy charged particles (enter and exit perpendicular to PQ) . With a field of B tesla it is found that the filter transmits α particle each of energy 2.2 MeV. The magnetic field is increased to 2.13B tesla and deuteron ions are passed into the filter. What is the approximate energy (In MeV) of each deuteron ionstransmitted by the filter?



Space for rough work

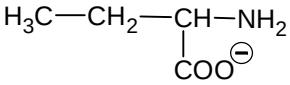
Chemistry PART - II

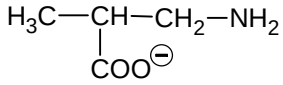
SECTION – A

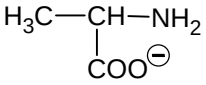
Single Correct Choice Type

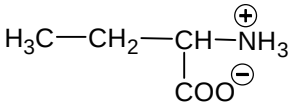
*This section contains **10 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.*

1. In alkaline medium alanine exits predominantly as

(A) 

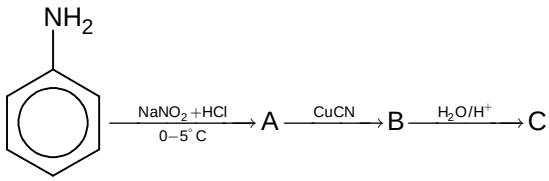
(B) 

(C) 

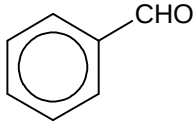
(D) 

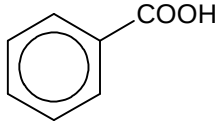
2. The Cannizzaro reaction is not given by:

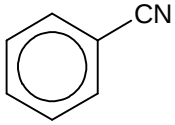
(A) trimethyl acetaldehyde (B) benzaldehyde
(C) acetaldehyde (D) formaldehyde

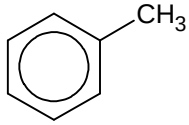
3. 

(C) is

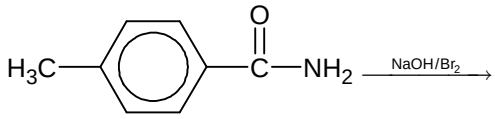
(A) 

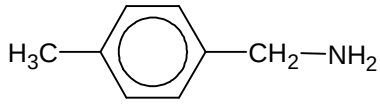
(B) 

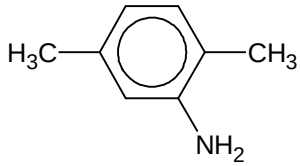
(C) 

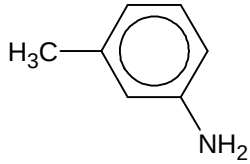
(D) 

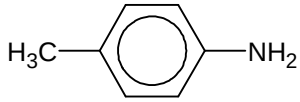
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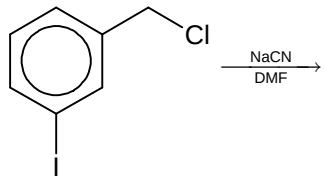
4. 

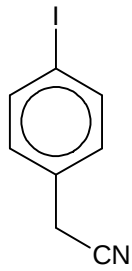
(A) 

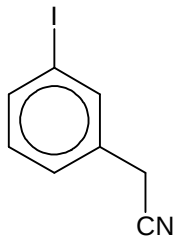
(B) 

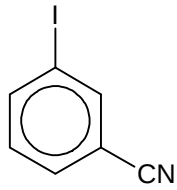
(C) 

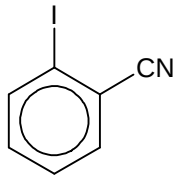
(D) 

5. 

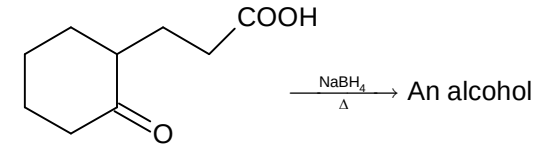
(A) 

(B) 

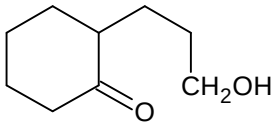
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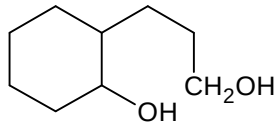
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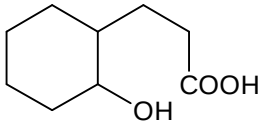
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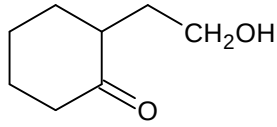
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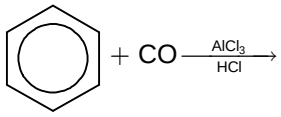
An alcohol is

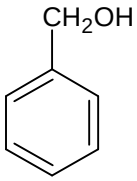
(A) 

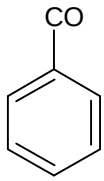
(B) 

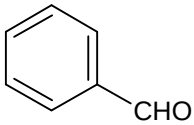
(C) 

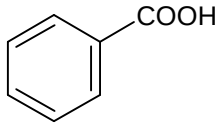
(D) 

7. 

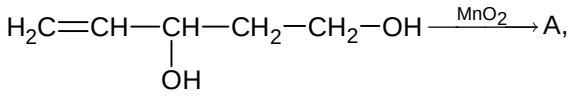
(A) 

(B) 

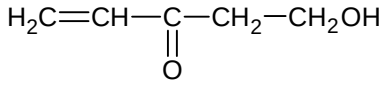
(C) 

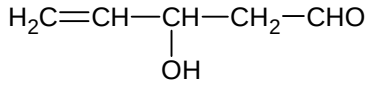
(D) 

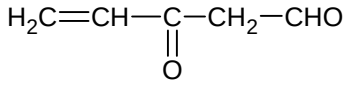
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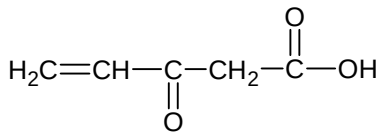
8. 

A is

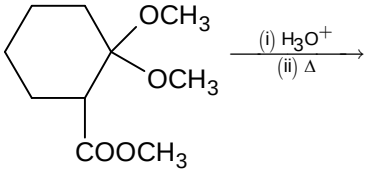
(A) 

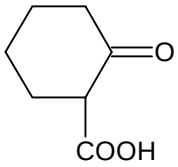
(B) 

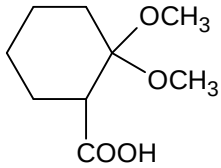
(C) 

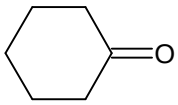
(D) 

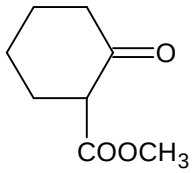
9. What is the final product of the following reaction?



(A) 

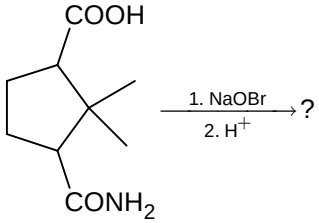
(B) 

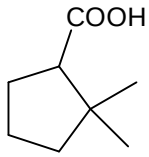
(C) 

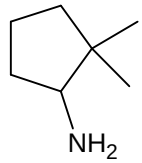
(D) 

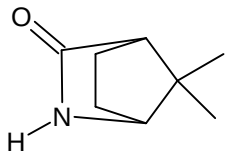
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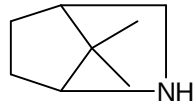
10. What is the final product of the following reaction?



(A) 

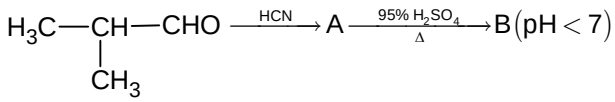
(B) 

(C) 

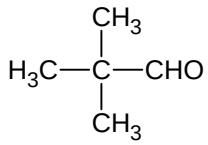
(D) 

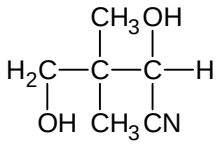
Multiple Correct Answer(s) Type

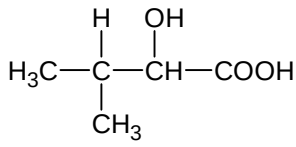
*This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.*

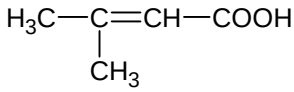
11. 

Which of these substances appear in reaction sequence?

(A) 

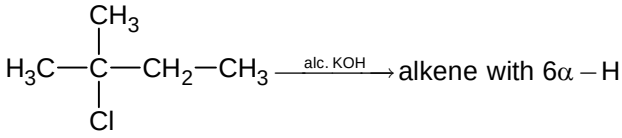
(B) 

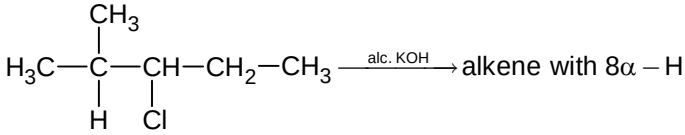
(C) 

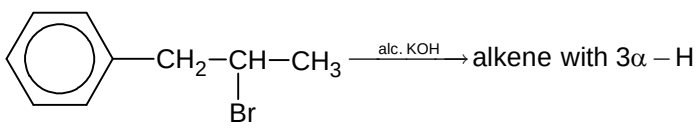
(D) 

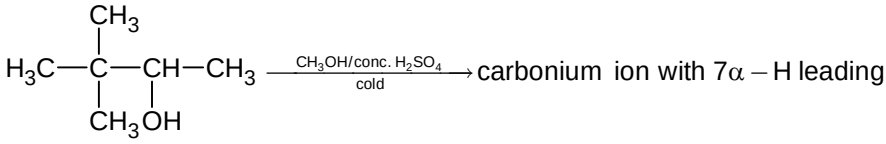
Space for rough work

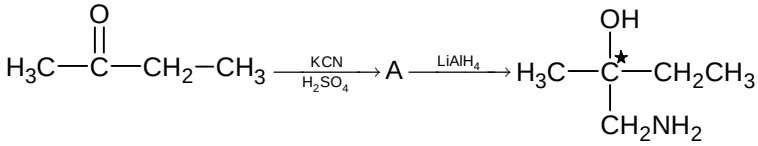
12. Which of the following statement(s) are correct?

(A) 

(B) 

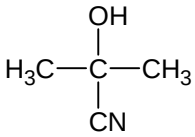
(C) 

(D) 

13. 

Which of these statements are correct?

(A) One of these chemical substances gives iodoform test

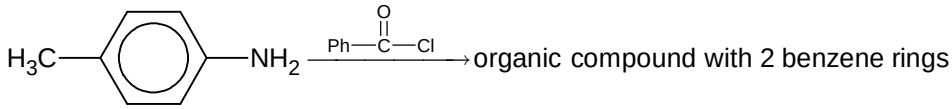
(B) 

is formed

(C) The final mixture is optically active

(D) The final mixture is racemic

Space for rough work

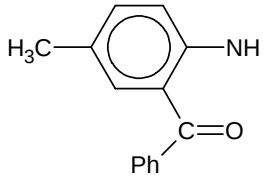
14. 

Which of the following statements is/are correct?

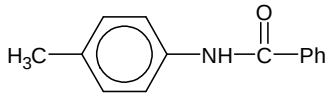
(A) One of the benzene rings is ortho & para directing

(B) One of the benzene rings is meta directing

(C) Organic compound is



(D) Organic compound is



15. Select the correct statement(s), among the following given statements

(A) $CH_3CH_2CH_2I$ will react more readily than $(CH_3)_2CHI$ for SN^2 reactions

(B) $CH_3CH_2CH_2Cl$ will react more readily than $CH_3CH_2CH_2Br$ for SN^2 reactions

(C) $CH_3CH_2CH_2CH_2Br$ will react more readily than $(CH_3)_3C - CH_2Br$ for SN^2 reactions

(D) $CH_3 - O - C_6H_5 - Br$ will react more readily than $NO_2 - C_6H_5 - CH_2Br$ for SN^2 reaction

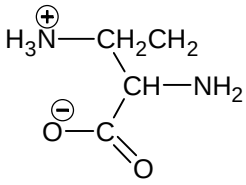
Space for rough work

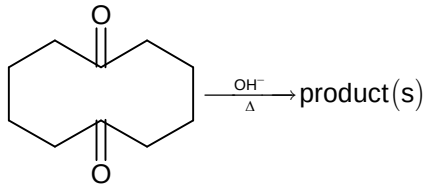
SECTION – C

(Integer Answer Type)

*This section contains **5** questions. The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The bubble corresponding to the correct is to be darkened in the ORS.*

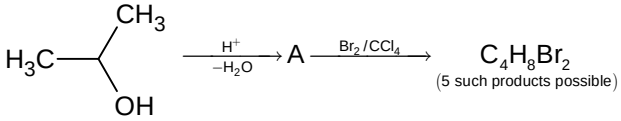
1. What is the total number of basic groups in the following?

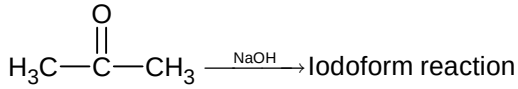


2. 

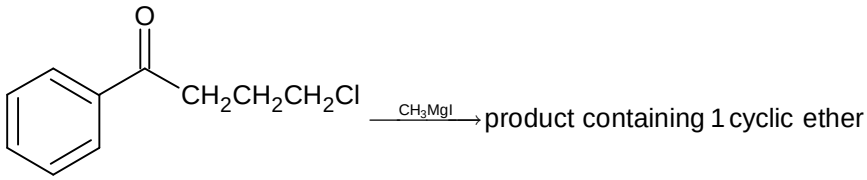
How many product(s) are possible?

3. How many structures of A are possible?



4. 

How many molecules of I_2 are required for 3 molecules of CH_3COCH_3?

5. 

How many C atoms are part of cyclic ether?

Space for rough work

Mathematics PART – III

SECTION – A

Single Correct Choice Type

*This section contains **10 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.*

1. If (a, 0); a > 0 is the point where the curve $y = \sin 2x - \sqrt{3}\sin x$ cuts the x-axis first, A is the area bounded by this part of the curve the origin and the positive x-axis, then
(A) 4A + 8 cos a = 7 (B) 4A + 8 sin a = 7
(C) 4A – 8 sin a = 7 (D) 4A – 8 cos a = 7

2. Let $f(x) = \lim_{h\to 0} \frac{\left(\sin(x+h)\right)^{\ln(x+h)} - (\sin x)^{\ln x}}{h}$ then $f\left(\frac{\pi}{2}\right)$ is
(A) equal to 0 (B) equal to 1
(C) $\ln\frac{\pi}{2}$ (D) none existent

3. For any natural number m, $\int \left(x^{7m} + x^{2m} + x^m\right)\left(2x^{6m} + 7x^m + 14\right)^{1/m} dx$ (where x > 0) equals
(A) $\frac{\left(7x^{7m} + 2x^{2m} + 14x^m\right)^{\frac{m+1}{m}}}{14(m+1)} + c$ (B) $\frac{\left(2x^{7m} + 14x^{2m} + 7x^m\right)^{\frac{m+1}{m}}}{14(m+1)} + c$
(C) $\frac{\left(2x^{7m} + 7x^{2m} + 14x^m\right)^{\frac{m+1}{m}}}{14(m+1)} + c$ (D) $\frac{\left(7x^{7m} + 2x^{2m} + x^m\right)^{\frac{m+1}{m}}}{14(m+1)} + c$

4. Let u(x) and v(x) are differentiable function such that $\frac{u(x)}{v(x)} = 7$. If $\frac{u'(x)}{v'(x)} = p$ and $\left(\frac{u(x)}{v(x)}\right)' = q$ then $\frac{p+q}{p-q}$ has the value equal to
(A) 1 (B) 0
(C) 7 (D) –7

Space for rough work

5. A function is represented parametrically by the equation $x = \frac{1+t}{t^3}$; $y = \frac{3}{2t^2} + \frac{2}{t}$ then $\frac{dy}{dx} - x\left(\frac{dy}{dx}\right)^3$ has the value equal to

(A) 2 (B) 0
(C) –1 (D) –2

6. Let g is the inverse function of f and $f'(x) = \frac{x^{10}}{(1+x^2)}$. If g(2) = a then g′(2) is equal to

(A) $\frac{5}{2^{10}}$ (B) $\frac{1+a^2}{a^{10}}$
(C) $\frac{a^{10}}{1+a^2}$ (D) $\frac{1+a^{10}}{a^2}$

7. Let $f:\left[0, \frac{\pi}{2}\right] \to R$ be continuous and satisfy $f'(x) = \frac{1}{1+\cos x}$ for all $x \in \left(0, \frac{\pi}{2}\right)$. If f(0) = 3 then $f\left(\frac{\pi}{2}\right)$ has the value

(A) $\frac{13}{4}$ (B) 2
(C) 4 (D) none of these

8. For the curve represented implicitly as $3^x - 2^y = 1$, the value of $\lim_{x\to\infty}\left(\frac{dy}{dx}\right)$ is

(A) equal to 1 (B) equal to 0
(C) equal to $\log_2 3$ (D) non existent

9. $\int(\sin(101x)\sin^{99} x)dx$ equals

(A) $\frac{\sin(100x)(\sin x)^{100}}{100} + c$ (B) $\frac{\cos(100x)(\sin x)^{100}}{100} + c$
(C) $\frac{\cos(100x)(\cos x)^{100}}{100} + c$ (D) $\frac{\sin(100x)(\sin x)^{101}}{101} + c$

Space for rough work

10. $\lim\limits_{x\to 0^+} \dfrac{1}{x\sqrt{x}}\left(a \arctan\dfrac{\sqrt{x}}{a} - b \arctan\dfrac{\sqrt{x}}{b}\right)$ the value equal to

(A) $\dfrac{a-b}{3}$ (B) 0

(C) $\dfrac{a^2-b^2}{6a^2b^2}$ (D) $\dfrac{a^2-b^2}{3a^2b^2}$

Multiple Correct Answer(s) Type

*This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.*

11. If $2^x + 2^y = 2^{x+y}$ then $\dfrac{dy}{dx}$ has the value equal to

(A) $-\dfrac{2^y}{2^x}$ (B) $\dfrac{1}{1-2^x}$

(C) $1-2^y$ (D) $\dfrac{2^x\left(1-2^y\right)}{2^y\left(2^x-1\right)}$

12. Two functions f and g have first and second derivative at x = 0 and satisfy the relations $f(0)=\dfrac{2}{g(0)}$; f′(0) = 2g′(0) = 4g(0); g″(0) = 5f″(0) = 6f(0) = 3 then

(A) if $h(x)=\dfrac{f(x)}{g(x)}$ then $h'(0)=\dfrac{15}{4}$ (B) if k(x) = f(x) g(x) sin x then k′(0) = 2

(C) $\lim\limits_{x\to 0}\dfrac{g'(x)}{f'(x)}=\dfrac{1}{2}$ (D) none of these

13. Let $f'(x)=3x^2\sin\dfrac{1}{x}-x\cos\dfrac{1}{x}$; if $x\neq 0$ f(0) = 0 and $f\left(\dfrac{1}{\pi}\right)=0$ then

(A) f(x) is continuous at x = 0 (B) f(x) is non derivable at x = 0

(C) f′(x) is continuous at x = 0 (D) f′(x) is non-derivable at x = 0

Space for rough work

14. Consider the function f(x) and g(x), both defined from $R \to R$ and are defined as $f(x) = 2x - x^2$ and $g(x) = x^n$ where $n \in N$. If the area between f(x) and g(x) is $\frac{1}{2}$ then n is a divisor of

(A) 12 (B) 15
(C) 20 (D) 30

15. Consider $f(x) = \begin{cases} \cos x & 0 \le x < \frac{\pi}{2} \\ \left(\frac{\pi}{2} - x\right)^2 & \frac{\pi}{2} \le x < \pi \end{cases}$ such that f is periodic with period x, then

(A) the range of f is $\left[0, \frac{\pi^2}{4}\right]$

(B) f is continuous for all real x, but not differentiable for some real x

(C) f is continuous for all real x

(D) the area bounded by y = f(x) and the x-axis from $x = -n\pi$ to $x = n\pi$ is $2n\left(1 + \frac{\pi^3}{24}\right)$ for $n \in N$

SECTION – C

(Integer Answer Type)

*This section contains **5** questions. The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The bubble corresponding to the correct is to be darkened in the ORS.*

1. Let $I_n = \int_0^{\pi/4} \tan^n x\,dx$ (n = 0, 1, 2, 3, ...) and $S_n = \sum_{n=0}^{n} \left(I_n I_{n+1} + I_n I_{n+3} + I_{n+1} I_{n+2}\right)$ then the value of $\lim_{x \to \infty} \left(S_n\right)$ is _____

Space for rough work

2. Let L denotes the value of $\lim_{x\to 0^+} \frac{1}{x}\cos^{-1}\left(\frac{\sin x}{x}\right)$ and U denotes the value of f″(0) where $f(x)=\int_0^x \sin\left(t^2-t+x\right)dt$, compute the value of $\frac{2U}{L^2}$ _____

3. Suppose $V=\int_0^{\pi/2} x\left|\sin^2 x-\frac{1}{2}\right|dx$, then the value of $\frac{32V}{\pi}$ _____

4. If the value of the integral $\int_0^{\pi}\frac{\cos 4x-\cos 4\alpha}{\cos x-\cos\alpha}dx$ is k($\pi\cos 2\alpha\cos\alpha$) then the value of k _____

5. A function f(x) is defined as $f(x)=\begin{cases} x^m\sin\frac{1}{x} & x\neq 0, m\in N \\ 0 & \text{if } x=0\end{cases}$. The least value of m for which f′(x) is continuous at x = 0 _____

Space for rough work